

VARshrink: An R Package for Shrinkage Estimation of High-Dimensional Vector Autoregressive Models

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Abstract Vector autoregressive (VAR) models are widely used to study dynamic relationships in multivariate time series, with applications ranging from economic and financial forecasting to gene network analysis in systems biology and brain connectivity analysis in neuroscience. High-dimensional VAR models involve many parameters, increasing computational costs and making coefficient estimation more challenging. Shrinkage methods address these challenges by regularizing the coefficient estimates and improving their stability. We present the R package VARshrink, which provides ridge, nonparametric, fully Bayesian, and semiparametric shrinkage estimators through a unified interface. As its main design feature, VARshrink returns objects that inherit from classes in the widely used `vars` package, allowing existing methods for analysis, visualization, and forecasting to be reused. The package also supports model comparison using information criteria adapted to shrinkage estimation. We illustrate the main workflow with reproducible R code and compare the implemented estimators in numerical experiments.

1 Introduction

Vector autoregressive (VAR) models describe the joint dynamics of multiple time series through their lagged values. They are widely used for macroeconomic forecasting and policy analysis, financial modeling, the inference of gene regulatory networks from time-course data in systems biology, and the analysis of dynamic brain connectivity in neuroscience (Hamilton, 1994; Koop and Korobilis, 2010; Opgen-Rhein and Strimmer, 2007; Harrison et al., 2003). However, the number of parameters increases rapidly with both the number of variables and the lag order. High-dimensional VAR models can therefore be computationally demanding and difficult to estimate reliably. Shrinkage methods address these challenges by regularizing coefficient estimates, thereby improving estimation stability and potentially forecasting performance.

Several R packages provide methods for VAR and related multivariate time-series models. The `MTS` package provides tools for VAR, VARMA, seasonal VARMA, and other multivariate time-series models (Tsay and Wood, 2018), while `vars` supports estimation, inference, forecasting, and diagnostic analysis for VAR, SVAR, and SVEC models (Pfaff and Stigler, 2018). For large systems, `BigVAR` estimates VAR and VARX models using structured penalties (Nicholson et al., 2026), whereas `bigtime` estimates sparse VAR, VARX, and VARMA models using structured lasso penalties (Wilms et al., 2023).

Several packages also support Bayesian VAR analysis. `bvartools` assists in setting up Bayesian inference for VAR and vector error-correction models (Mohr, 2024). `bayesianVARs` implements MCMC estimation of Bayesian VARs with shrinkage priors and stochastic volatility (Gruber et al., 2026), while `bvars` implements time-varying parameter VARs with stochastic volatility (Primiceri, 2005; Krueger, 2015). `BVAR` implements hierarchical prior selection for conjugate priors following Giannone et al. (2015; Kuschnig and Vashold, 2021). `bsvarSIGNs` uses Minnesota and dummy-observation priors with estimated hyperparameters in structural VARs identified by sign, zero, and narrative restrictions (Wang and Woźniak, 2025).

In this study, we focus on shrinkage estimation of VAR model parameters. Shrinkage estimation methods have played crucial roles in high-dimensional statistical modeling (Ledoit and Wolf, 2004; Böhm and von Sachs, 2009; Fiecas and Ombao, 2011; Beltrachini et al., 2013). For VAR models, an extensive literature has developed on regularization and shrinkage priors, including Minnesota-type priors and more recent global-local shrinkage priors. Frequentist penalized estimation has also been widely studied. Approaches directly related to this package include a Stein-type nonparametric shrinkage estimation method (Opgen-Rhein and Strimmer, 2007), Bayesian VARs using informative priors (Doan et al., 1984; Litterman, 1986; Bańbura et al., 2010; Koop and Korobilis, 2010), Bayesian VARs using noninformative priors (Sun and Ni, 2004; Ni and Sun, 2005), and a semiparametric Bayesian approach adopting a modified K-fold cross-validation (Lee et al., 2016).

Shrinkage methods developed for purposes other than multivariate time series analysis can also be applied to the estimation of VAR parameters. For instance, the function `cov.shrink()` in the package `corpcor` was developed to compute shrinkage estimates of covariances, but it has also been used to estimate VAR coefficients (Opgen-Rhein and Strimmer, 2007; Schäfer et al., 2017). Moreover, VAR models can be reformulated as multivariate regression problems, allowing penalized least squares

methods to be used for shrinkage estimation of VAR parameters. Examples include the functions `lm.gls()` for generalized least squares and `lm.ridge()` for ridge regression in the package **MASS** (Ripley et al., 2018); the function `glmnet()` for Lasso and Elastic-Net regularized generalized linear models in the package **glmnet** (Friedman et al., 2018); the function `linearRidge()` for ridge regression in the package **ridge** (Moritz and Cule, 2018).

While Bayesian approaches have been widely used in the literature, nonparametric and semiparametric approaches have advantages for high-dimensional VARs with several hundred or more time series variables due to their relatively low computational cost (Opgen-Rhein and Strimmer, 2007). In contrast, Bayesian approaches can flexibly impose proper assumptions on multivariate time series data, such as VAR roots near unity and correlations among noise processes (Lee et al., 2016). In this sense, a semiparametric approach serves as a trade-off between nonparametric and parametric approaches (Lee et al., 2016).

The main design contribution of **VARshrink** is to extend the workflow of **vars** from ordinary least squares to shrinkage estimation. Users can apply familiar methods from **vars**, including ARCH-LM tests, causality analysis, diagnostic plots, forecasting, impulse response analysis, and forecast error variance decomposition, directly to **R** objects obtained via shrinkage estimation methods in **VARshrink**. This compatibility also allows shrinkage methods to be introduced within established applied and teaching workflows based on **vars**.

Relative to existing implementations, **VARshrink** brings together ridge regression, a Stein-type nonparametric estimator, fully Bayesian estimation, and semiparametric Bayesian estimation in one interface. Its fully Bayesian method includes both a natural conjugate normal-Wishart prior and a non-conjugate shrinkage-reference prior, with scale-mixture likelihoods available for robust estimation. The package also provides likelihood and information criterion methods that account for the selected noise distribution and the effective number of parameters. These features are intended to complement packages specializing in structured penalties, hierarchical Bayesian prior selection, stochastic volatility, or structural identification.

This paper is organized as follows. In Section 2, we formulate VAR models as a multivariate regression problem, which simplifies the implementation of the package. We also describe the general distributional assumptions for the noise term within a Bayesian framework. Closed-form expressions for the shrinkage estimators implemented in the package are presented to clarify the role of the shrinkage intensity parameters in each method. In addition, we explain how the effective number of parameters is computed for shrinkage estimators. Section 3 introduces the common interface function and provides sample code snippets to demonstrate the usefulness of the package. In Section 4, we present numerical experiments using benchmark and simulated data to compare the performance of the shrinkage estimation methods. Discussion and conclusions are given in Sections 5 and 6.

2 Models

2.1 Multivariate linear regression formulation

Let $\mathbf{y}_t = (y_{t1}, y_{t2}, \dots, y_{tK})^\top$ denote a $K \times 1$ vector of endogenous variables. A VAR model of order p can be expressed as

$$\mathbf{y}_t = \mathbf{A}_1 \mathbf{y}_{t-1} + \dots + \mathbf{A}_p \mathbf{y}_{t-p} + \mathbf{C} \mathbf{d}_t + \boldsymbol{\epsilon}_t, \quad (1)$$

where $\mathbf{d}_t$ is an $L \times 1$ vector of deterministic regressors, $\boldsymbol{\epsilon}_t$ is a $K \times 1$ noise vector, and $\mathbf{A}_1, \dots, \mathbf{A}_p$ and $\mathbf{C}$ are coefficient matrices (Hamilton, 1994; Tsay, 2005).

The model equation (1) can be expressed in the matrix form of a multivariate regression as

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\Psi} + \mathbf{E} \in \mathbb{R}^{N \times K}, \quad (2)$$

where $\boldsymbol{\Psi} = (\mathbf{A}_1, \mathbf{A}_2, \dots, \mathbf{A}_p, \mathbf{C})^\top$ is a $(Kp + L) \times K$ matrix of coefficients,

$$\begin{aligned} \mathbf{Y} &= (\mathbf{y}_{p+1}, \mathbf{y}_{p+2}, \dots, \mathbf{y}_T)^\top \in \mathbb{R}^{N \times K}, \\ \mathbf{X} &= (\mathbf{x}_{p+1}, \mathbf{x}_{p+2}, \dots, \mathbf{x}_T)^\top \in \mathbb{R}^{N \times (Kp+L)} \end{aligned} \quad (3)$$

are data matrices with $\mathbf{x}_t = (\mathbf{y}_{t-1}^\top, \dots, \mathbf{y}_{t-p}^\top, \mathbf{d}_t^\top)^\top \in \mathbb{R}^{Kp+L}$, $\mathbf{E} = (\boldsymbol{\epsilon}_{p+1}, \dots, \boldsymbol{\epsilon}_T)^\top$, and $N = T - p$.

2.2 Mathematical expressions for shrinkage estimators

In version 0.5 of the **R** package **VARshrink**, ordinary least squares (OLS) and five shrinkage estimation methods are implemented: multivariate ridge regression (Hoerl and Kennard, 1970; Golub et al.,

1979; Ripley et al., 2018), a Stein-type nonparametric shrinkage method (Schäfer and Strimmer, 2005; Opgen-Rhein and Strimmer, 2007; Schäfer et al., 2017), a full Bayesian shrinkage method (Sun and Ni, 2004; Ni and Sun, 2005), and two semiparametric Bayesian shrinkage methods based on parameterized and K-fold cross-validation, respectively (Lee et al., 2016). OLS is included as a baseline for settings in which the VAR is estimable without shrinkage. For each shrinkage estimation method, the shrinkage estimator of the coefficient matrix Ψ takes a distinct mathematical form.

(1) Multivariate ridge regression

The ridge regression estimator, $\hat{\Psi}^R(\lambda)$, can be expressed in a closed form as follows (Hoerl and Kennard, 1970):

$$\hat{\Psi}^R(\lambda) = (\mathbf{X}^\top \mathbf{X} + N\lambda \mathbf{I})^{-1} \mathbf{X}^\top \mathbf{Y}, \quad (4)$$

where $\lambda \geq 0$ is called the regularization parameter or the shrinkage parameter.

(2) Stein-type nonparametric shrinkage method

Suppose that the VAR model does not include a constant term, that is, a regressor whose value remains constant over time (e.g., $d_t = 1$). Assuming that the data matrices $\mathbf{X}$ and $\mathbf{Y}$ have been mean-corrected, let $\mathbf{S}_{XX} = (N-1)^{-1} \mathbf{X}^\top \mathbf{X}$, $\mathbf{S}_{XY} = (N-1)^{-1} \mathbf{X}^\top \mathbf{Y}$, and $\mathbf{S}_{YY} = (N-1)^{-1} \mathbf{Y}^\top \mathbf{Y}$ denote the sample covariance matrices, $\mathbf{D}_X = \text{diag}(\mathbf{S}_{XX})$ and $\mathbf{D}_Y = \text{diag}(\mathbf{S}_{YY})$ denote the diagonal matrices of sample variances, and $\mathbf{R}_{XX}$, $\mathbf{R}_{XY}$, and $\mathbf{R}_{YY}$ denote the sample correlation matrices. The shrinkage estimators of the correlation matrices are defined as:

$$\hat{\mathbf{R}}_{XX} = (1 - \lambda) \mathbf{R}_{XX} + \lambda \mathbf{I}, \quad \hat{\mathbf{R}}_{XY} = (1 - \lambda) \mathbf{R}_{XY}, \quad (5)$$

where $0 \leq \lambda \leq 1$ is the shrinkage parameter for the correlations. And the shrinkage estimators of the variance matrices are defined as:

$$\hat{\mathbf{D}}_X = (1 - \lambda_v) \mathbf{D}_X + \lambda_v s_{\text{med}} \mathbf{I}, \quad \hat{\mathbf{D}}_Y = (1 - \lambda_v) \mathbf{D}_Y + \lambda_v s_{\text{med}} \mathbf{I}, \quad (6)$$

where $0 \leq \lambda_v \leq 1$ is the shrinkage parameter for the variances, and s_{med} is the median of all sample variances.

The nonparametric shrinkage (NS) estimator of Ψ can be expressed as follows (Schäfer and Strimmer, 2005; Opgen-Rhein and Strimmer, 2007):

$$\hat{\Psi}^N(\lambda, \lambda_v) = \hat{\mathbf{D}}_X^{-1/2} \hat{\mathbf{R}}_{XX}^{-1} \hat{\mathbf{R}}_{XY} \hat{\mathbf{D}}_Y^{1/2}. \quad (7)$$

The shrinkage estimate $\hat{\Psi}^N(\lambda, \lambda_v)$ approaches the ordinary least squares estimate, $\hat{\Psi}^{\text{OLS}} = \mathbf{S}_{XX}^{-1} \mathbf{S}_{XY}$, as λ and λ_v approach zero.

(3) Full Bayesian shrinkage method

The full Bayesian shrinkage method supports several likelihood and prior specifications.

Likelihoods We assume that the noise vectors are independent and identically distributed from a scale mixture of multivariate normal distributions, including multivariate normal distributions, multivariate t-distributions (Ni and Sun, 2005), and multivariate Laplace distributions (MLD) (Eltoft et al., 2006). By incorporating scale mixture distributions in the model assumption, we can handle outliers appropriately and produce robust estimates (West, 1984). In specific, we consider that a noise vector can be expressed as a product

$$\epsilon_t = \mathbf{z}_t q_t^{-1/2}, \quad (8)$$

where $\mathbf{z}_t \sim N_K(\mathbf{0}, \Sigma)$ is a $K \times 1$ vector having a multivariate normal distribution with the mean vector of zeros and the covariance matrix of Σ , and $q_t \sim \text{Gamma}(\nu/2, \nu/2)$ is a random variable having a gamma distribution with shape $\alpha = \nu/2$ and rate $\beta = \nu/2$. The ϵ has a multivariate t-distribution with degrees of freedom ν when $1 < \nu < \infty$, a multivariate normal distribution as $\nu \rightarrow \infty$, and a multivariate Cauchy distribution when $\nu = 1$. More detailed discussions can be found in the package vignette and Lee et al. (2016).

Conjugate prior Let $\psi = \text{vec}(\Psi) \in \mathbb{R}^J$ denote the vectorized form of $\Psi \in \mathbb{R}^{(Kp+L) \times K}$, where $J = K(Kp+L)$. The natural conjugate prior implemented in **VARshrink** is a normal-Wishart prior.

Conditional on Σ , the prior for ψ is

$$(\psi|\Sigma, \delta) \sim N_J(\mathbf{0}, \delta \Sigma \otimes \mathbf{I}), \quad (9)$$

where $\otimes$ represents the Kronecker product of two matrices and $\delta \equiv 1/\lambda > 0$ is a reciprocal of a shrinkage parameter $\lambda > 0$, and an inverse Wishart distribution with a positive definite scale matrix $\mathbf{L}_0$ and degrees of freedom $m_0 > K - 1$ for Σ as

$$(\Sigma|\mathbf{L}_0, m_0) \sim \text{InvWishart}(\mathbf{L}_0, m_0). \quad (10)$$

The prior distributions for δ and $w = \nu/2$ are given by

$$\begin{aligned} \pi(\delta) &\propto 1, \\ w &\sim \text{Gamma}(a_0, b_0), \end{aligned} \quad (11)$$

for some $a_0, b_0 > 0$.

VARshrink uses a Gibbs MCMC method to sample $(\psi, \mathbf{Q}, \Sigma, w, \delta)$ from the conditional posterior distributions; see the package vignette for more details. Let $\mathbf{Q} = \text{diag}(q_{p+1}, \dots, q_T)$. With $\lambda \equiv 1/\delta$, the mean of the conditional posterior distribution of ψ given $(\mathbf{Q}, \Sigma, w, \delta; \mathbf{Y})$ can be written as

$$\hat{\Psi}_{\text{CJ}}^{\text{F}}(\lambda) = (\mathbf{X}^{\top} \mathbf{Q} \mathbf{X} + \lambda \mathbf{I})^{-1} \mathbf{X}^{\top} \mathbf{Q} \mathbf{Y}. \quad (12)$$

Non-conjugate priors The non-conjugate prior adopted in **VARshrink** is the shrinkage-reference prior of [Ni and Sun \(2005\)](#), which is given by

$$\begin{aligned} (\psi|\delta) &\sim N_J(\mathbf{0}, \delta \mathbf{I}), \\ \pi(\Sigma) &\propto |\Sigma|^{-1} \prod_{1 \leq i < j \leq K} (\lambda_i - \lambda_j)^{-1}, \\ \pi(\delta) &\propto 1, \\ w &\sim \text{Gamma}(a_0, b_0), \end{aligned} \quad (13)$$

where $\lambda_1 > \lambda_2 > \dots > \lambda_K$ are eigenvalues of Σ , $w = \nu/2$, and $a_0, b_0 > 0$.

With $\lambda \equiv 1/\delta$, the mean of the conditional posterior distribution of ψ given $(\mathbf{Q}, \Sigma, w, \delta; \mathbf{Y})$ is given by

$$\hat{\psi}_{\text{NCJ}}^{\text{F}}(\lambda) = (\Sigma^{-1} \otimes (\mathbf{X}^{\top} \mathbf{Q} \mathbf{X}) + \lambda \mathbf{I}_J)^{-1} \text{vec}(\mathbf{X}^{\top} \mathbf{Q} \mathbf{Y} \Sigma^{-1}). \quad (14)$$

(4) Semi-parametric shrinkage method

In contrast to the full Bayesian approach, the semiparametric Bayesian approach selects the shrinkage parameter $0 \leq \lambda < 1$ through parameterized cross-validation ([Lee et al., 2016](#)). Once the λ is fixed, the parameters ψ , Σ , and $\mathbf{Q}$ are estimated at the mode of the marginal posterior density $\pi(\psi, \Sigma|\lambda; \mathbf{Y})$. In the case of a non-conjugate prior for ψ ([Lee et al., 2016](#)), the mode $\hat{\psi}^{\text{S}}(\lambda)$ is given by

$$\hat{\psi}^{\text{S}}(\lambda) = \left[(\Sigma^{-1} \otimes (\mathbf{X}^{\top} \mathbf{Q} \mathbf{X})) + \frac{(N-1)\lambda}{1-\lambda} \mathbf{I}_J \right]^{-1} \text{vec}(\mathbf{X}^{\top} \mathbf{Q} \mathbf{Y} \Sigma^{-1}). \quad (15)$$

The covariance matrix Σ is estimated in the same manner.

2.3 Estimation of the effective number of parameters

Although all shrinkage estimators are expressed in slightly different forms, they approach the ordinary least squares estimator (or the maximum likelihood estimator) as the shrinkage parameters decrease to zero, and they are shrunk toward zeros as the shrinkage parameters increase. It implies that the effective number of parameters underlying a shrinkage estimator is controlled by the shrinkage parameter values.

The shrinkage estimators presented in the previous section can be expressed in a general form:

$$\hat{\psi}(\lambda_0) = \left[(\Sigma^{-1} \otimes (\mathbf{X}^{\top} \mathbf{Q} \mathbf{X})) + \lambda_0 \mathbf{I}_J \right]^{-1} \text{vec}(\mathbf{X}^{\top} \mathbf{Q} \mathbf{Y} \Sigma^{-1}), \quad (16)$$

where $\lambda_0 > 0$ is the shrinkage parameter. A noise covariance matrix is approximated by its diagonal

elements as $\Sigma \approx \mathbf{D}_e \equiv \text{diag}(\Sigma) = \text{diag}(\sigma_{11}, \dots, \sigma_{KK})$ to simplify the expression for $\hat{\Psi}(\lambda_0)$ as:

$$\hat{\Psi}(\lambda_0) \approx \left[\left(\mathbf{I}_K \otimes (\mathbf{X}^\top \mathbf{Q} \mathbf{X}) \right) + \lambda_0 \mathbf{D}_e \otimes \mathbf{I}_{J/K} \right]^{-1} \text{vec}(\mathbf{X}^\top \mathbf{Q} \mathbf{Y}),$$

or equivalently, for each $j = 1, 2, \dots, K$,

$$\hat{\Psi}_j(\lambda_0) \approx \left(\mathbf{X}^\top \mathbf{Q} \mathbf{X} + \lambda_0 \sigma_{jj} \mathbf{I}_{J/K} \right)^{-1} \mathbf{X}^\top \mathbf{Q} \mathbf{y}_j, \quad (17)$$

where $\hat{\Psi}_j(\lambda_0)$ and $\mathbf{y}_j$ are the j th column vectors of $\hat{\Psi}(\lambda_0)$ and $\mathbf{Y}$.

From the definition of a hat matrix, $\mathbf{H}(j, \lambda_0)$, we have $\hat{\mathbf{y}}_j \equiv \mathbf{H}(j, \lambda_0) \mathbf{y}_j = \mathbf{X} \hat{\Psi}_j(\lambda_0)$, and the hat matrix can be expressed as: $\mathbf{H}(j, \lambda_0) \equiv \mathbf{X} \left(\mathbf{X}^\top \mathbf{Q} \mathbf{X} + \lambda_0 \sigma_{jj} \mathbf{I}_{J/K} \right)^{-1} \mathbf{X}^\top \mathbf{Q}$. Let $\mathbf{X}^\top \mathbf{Q} \mathbf{X} = \mathbf{P} \mathbf{L} \mathbf{P}^\top$ denote the eigenvalue decomposition of $\mathbf{X}^\top \mathbf{Q} \mathbf{X}$, where $\mathbf{L} = \text{diag}(l_{11}, l_{22}, \dots)$ is the diagonal matrix of eigenvalues. The effective number of parameters, κ_{eff} , is computed by the trace of the hat matrix as:

$$\kappa_{\text{eff}}(\lambda_0) = \sum_{j=1}^K \text{tr}(\mathbf{H}(j, \lambda_0)) = \sum_{j=1}^K \text{tr} \left(\left(\mathbf{L} + \lambda_0 \sigma_{jj} \mathbf{I}_{J/K} \right)^{-1} \mathbf{L} \right) = \sum_{j=1}^K \sum_{i=1}^{J/K} \frac{l_{ii}}{l_{ii} + \lambda_0 \sigma_{jj}}. \quad (18)$$

With the estimated value for the effective number of parameters, the AIC and BIC values can be calculated as:

$$\begin{aligned} \text{AIC}(\lambda_0) &= -2\log\text{Lik}(\lambda_0) + 2\kappa_{\text{eff}}(\lambda_0), \\ \text{BIC}(\lambda_0) &= -2\log\text{Lik}(\lambda_0) + \kappa_{\text{eff}}(\lambda_0) \log(N), \end{aligned} \quad (19)$$

where $\log\text{Lik}$ represents the log-likelihood. We can use the AIC and BIC values to select an optimal value of the lag order, p .

3 Overview of the package

The **R** package **VARshrink** performs shrinkage estimation of the VAR model coefficients Ψ in (2) from multivariate time series data $\mathbf{y}_1, \mathbf{y}_2, \dots, \mathbf{y}_T \in \mathbb{R}^K$. It estimates the so-called shrinkage intensity parameter $\lambda > 0$, whose interpretation slightly differs across shrinkage estimation methods. Depending on the method, additional model parameters may also be estimated, such as the $K \times K$ covariance matrix Σ of the noise vector $\epsilon_t \in \mathbb{R}^K$ in (8).

3.1 Package architecture

The software architecture of **VARshrink** is illustrated in Figure 1. The main interface function, `VARshrink()`, receives multivariate time series data in a $T \times K$ matrix form, and outputs the estimated parameters as an instance of the `varshrinkest` class. An instance of this class can then be used to produce useful time series analysis results through various class methods and analysis functions. Table 1 summarizes the structure of **VARshrink**, listing the available class methods and analysis functions.

The class `varshrinkest` inherits from the class `varest` in the **vars** package. Consequently, all class methods available for `varest` can also be applied to `varshrinkest`, such as `fevd()`, `Phi()`, and `plot()`. We also note that the classes `varshrinkest`, `varshirf`, and `varshsum` inherit from `varest`, `varirf`, and `varsum`, respectively, in the **vars** package.

The class method `irf.varshrinkest()` retains the interface and identification options of `vars::irf()`. For shrinkage estimators, confidence intervals are obtained via bootstrap procedures that require repeatedly refitting the model. In contrast, when the full Bayesian method is fitted with `store_mcmc = TRUE`, `irf()` can construct Bayesian credible intervals directly from the stored MCMC draws of the VAR coefficients, avoiding the computational cost of bootstrap-based interval estimation.

3.2 Main function

We provide a common **R** function interface `VARshrink()` for running the estimation methods, which is defined by

```
VARshrink(y, p = 1, type = c("const", "trend", "both", "none"),
          season = NULL, exogen = NULL, method = c("ridge", "ns",
          "fbayes", "sbayes", "kcv", "ols"), lambda = NULL, lambda_var = NULL,
```

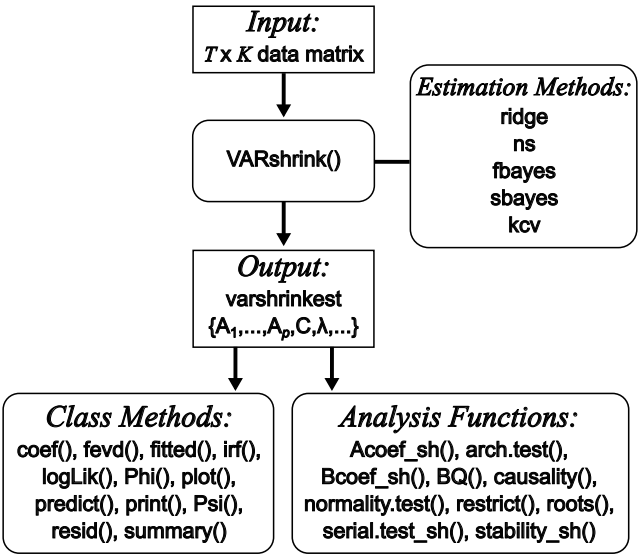


Figure 1: Systematic diagram of the VARshrink package.

Table 1: Structure of the VARshrink package.

Function or method	Class	Class methods	Analysis functions for class
VARshrink	varshrinkest, varest	coef, fevd, fitted, irf, logLik, Phi, plot, predict, print, Psi, resid, summary	Acoef_sh, arch.test, Bcoef_sh, BQ, causality, normality.test, restrict, roots, serial.test_sh, stability_sh
fevd	varfevd	plot, print	
irf	varshirf, varirf	plot, print	
predict	varprd	plot, print	fanchart
summary	varshsum, varsum	print	
arch.test	varcheck	plot, print	
normality.test	varcheck	plot, print	
serial.test_sh	varcheck	plot, print	
stability_sh	varstabil	plot, print	

dof = Inf, ...)

The input arguments are described as follows.

- y : $A \times T \times K$ matrix of endogenous variables
- p : Integer for the lag order
- type: Type of deterministic regressors to include.
 1. "const" - the constant: $\mathbf{d}_t = 1$.
 2. "trend" - the trend: $\mathbf{d}_t = t$.
 3. "both" - both the constant and the trend: $\mathbf{d}_t = (t, 1)^\top$.
 4. "none" - no deterministic regressors.
- season: An integer value of frequency for inclusion of centered seasonal dummy variables.
- exogen: $A \times T \times L$ matrix of exogenous variables.
- method:
 1. "ridge" - multivariate ridge regression (Hoerl and Kennard, 1970; Golub et al., 1979; Ripley et al., 2018).
 2. "ns" - a Stein-type nonparametric shrinkage method (Schäfer and Strimmer, 2005; Opgen-Rhein and Strimmer, 2007; Schäfer et al., 2017).
 3. "fbayes" - a full Bayesian shrinkage method using noninformative priors (Sun and Ni, 2004; Ni and Sun, 2005).
 4. "sbayes" - a semiparametric Bayesian shrinkage method using parameterized cross-validation (Lee et al., 2016).
 5. "kcv" - a semiparametric Bayesian shrinkage method using K-fold cross-validation (Lee et al., 2016).
 6. "ols" - ordinary least squares estimation, calling `vars::VAR()`.
- lambda, lambda_var: Shrinkage parameter value(s). Use of this parameter is slightly different for each method: the same value does not imply the same shrinkage estimates. See the description in the previous section for the use of shrinkage parameters in each method.
- dof: Degree of freedom of multivariate t-distribution for noise. Valid only for method = "fbayes" and method = "sbayes". dof = Inf means multivariate normal distribution.
- ...: Additional method-specific options, including `store_mcmc = TRUE` for retaining MCMC draws from the full Bayesian method.

3.3 Sample code snippets analysis

Simulated time series data of length $T = 100$ were generated from a VAR model with order $p = 1$ and dimension $K = 2$, assuming multivariate normal noise. The model parameters were set as $\mathbf{A}_1 = 0.5\mathbf{I}_2$, $\mathbf{C} = (0.2, 0.7)^\top$, and $\Sigma = 0.1^2\mathbf{I}_2$, as follows:

```
set.seed(1000)
myCoef <- list(A = list(matrix(c(0.5, 0, 0, 0.5), 2, 2)), c = c(0.2, 0.7))
myModel <- list(Coef = myCoef, Sigma = diag(0.1^2, 2), dof = Inf)
Y <- simVARmodel(numT = 100, model = myModel, burnin = 10)
resu_estim <- list()
```

The multivariate ridge regression for VAR models can be performed as follows. The output displays all the candidate lambda values along with their corresponding generalized cross-validation (GCV) scores (Golub et al., 1979). The VAR parameters are then estimated using the lambda value that yields the minimum GCV score.

```
resu_estim$`Ridge regression` <-
  VARshrink(Y, p = 1, type = "const", method = "ridge", lambda = NULL)
resu_estim$`Ridge regression`

#>
#> VAR Shrinkage Estimation Results:
#> =====
#>
```

```

#> Estimated coefficients for equation y1:
#> =====
#> Call:
#> y1 = y1.l1 + y2.l1 + const
#>
#>      y1.l1      y2.l1      const
#> 0.61424699 0.07318362 0.05264740
#>
#>
#> Estimated coefficients for equation y2:
#> =====
#> Call:
#> y2 = y1.l1 + y2.l1 + const
#>
#>      y1.l1      y2.l1      const
#> 0.1195550 0.3254618 0.9055450
#>
#>
#> lambda: 5e-04 (estimated: TRUE)
#> GCV: 0.0186

```

The `summary()` method reports the estimated coefficient matrices, shrinkage parameters, noise covariance matrix, and method-specific fitting information.

After estimation, the fitted object can be used in a standard VAR workflow for diagnostic checking, forecasting, impulse response analysis, and forecast error variance decomposition. The returned objects have their own `plot()` methods.

```

fit_ridge <- resu_estim$`Ridge regression`

plot(fit_ridge)

diagnostics <- list(
  serial = serial.test_sh(fit_ridge),
  arch = arch.test(fit_ridge),
  normality = normality.test(fit_ridge),
  stability = stability_sh(fit_ridge)
)

plot(predict(fit_ridge, n.ahead = 5), names = "y1")

plot(irf(fit_ridge, impulse = "y1", response = "y2", n.ahead = 10,
  ortho = TRUE, boot = TRUE))

plot(fevd(fit_ridge, n.ahead = 10))

```

The class methods `AIC()` and `BIC()` can be used to compute the Akaike Information Criterion (AIC) and the Bayesian Information Criterion (BIC) as follows:

```

c(AIC = AIC(resu_estim$`Ridge regression`),
  BIC = BIC(resu_estim$`Ridge regression`))

#>      AIC      BIC
#> -366.2358 -351.5082

```

An OLS and the other shrinkage estimation methods ("ns", "fbayes", "sbayes", and "kcv") can be run similarly. For "ns", "sbayes", and "kcv", if the input arguments `lambda` and `lambda_var` are set to NULL, the shrinkage parameters λ and λ_v are determined automatically. For "fbayes", "sbayes", and "kcv", the degree of freedom ν of the multivariate t -distribution for noise can either be set to a fixed value or estimated automatically by specifying the argument `dof`. For "fbayes", "sbayes", and "kcv", `prior_type` can be set to either "CJ" or "NCJ" corresponding to conjugate and non-conjugate prior distributions, respectively.

```
resu_estim$`Ordinary least squares` <-
```

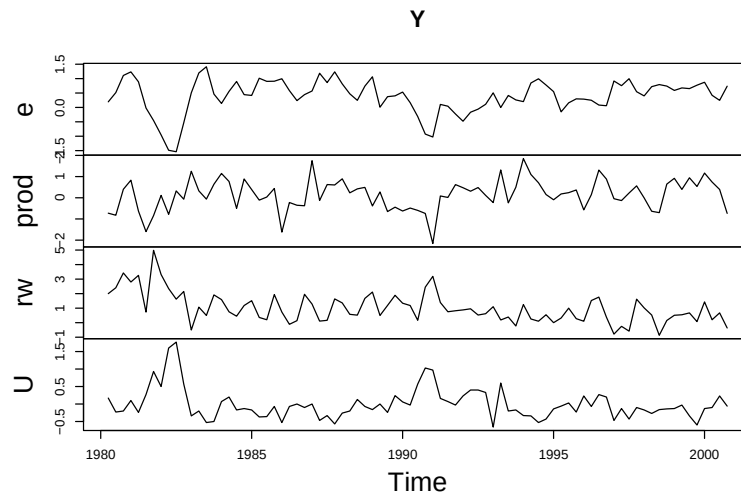


Figure 2: Benchmark dataset obtained by differencing the Canada time series from the vars package.

```
VARshrink(Y, p = 1, type = "const", method = "ols")
resu_estim$`Nonparametric shrinkage` <-
  VARshrink(Y, p = 1, type = "none", method = "ns", lambda = NULL,
    lambda_var = NULL)
resu_estim$`Full Bayes (fixed dof)` <-
  VARshrink(Y, p = 1, type = "const", method = "fbayes", dof = 6,
    prior_type = "NCJ", burnincycle = 1000, mcmccycle = 2000,
    store_mcmc = TRUE)
resu_estim$`Full Bayes (estim dof)` <-
  VARshrink(Y, p = 1, type = "const", method = "fbayes", dof = NULL,
    prior_type = "NCJ", burnincycle = 1000, mcmccycle = 2000,
    store_mcmc = TRUE)
resu_estim$`Semi Bayes (fixed dof)` <-
  VARshrink(Y, p = 1, type = "const", method = "sbayes", dof = 6,
    prior_type = "NCJ", lambda = NULL, lambda_var = NULL, num_folds = 5)
resu_estim$`Semi Bayes (estim dof)` <-
  VARshrink(Y, p = 1, type = "const", method = "sbayes", dof = NULL,
    prior_type = "NCJ", lambda = NULL, lambda_var = NULL, num_folds = 5)
resu_estim$`K-fold CV (fixed dof)` <-
  VARshrink(Y, p = 1, type = "const", method = "kcv", dof = 6,
    prior_type = "NCJ", lambda = NULL, lambda_var = NULL, num_folds = 5)
resu_estim$`K-fold CV (estim dof)` <-
  VARshrink(Y, p = 1, type = "const", method = "kcv", dof = NULL,
    prior_type = "NCJ", lambda = NULL, lambda_var = NULL, num_folds = 5)
```

4 Illustrative examples

4.1 Benchmark data

The Canada dataset is a benchmark macroeconomic dataset included in the `vars` package. It contains four time series variables: employment (`e`), labor productivity (`prod`), real wage (`rw`), and unemployment rate (`U`), with 84 observations. We differenced the data to remove the trend, resulting in $T = 83$. Figure 2 shows the differenced series.

We illustrate the use of AIC and BIC for comparing VAR models with varying lag order $p = 1, 2, 3$. OLS is included as a baseline, shrinkage parameters were selected automatically for the shrinkage methods, and for “fbayes”, “sbayes”, and “kcv”, the `dof` argument was set to `NULL`. Table 2 shows that, using the NS method, AIC attains its minimum at $p = 3$, whereas BIC attains its minimum at $p = 2$.

The parameters estimated by the NS method with $p = 2$ can be further analyzed using the class methods and functions listed in Table 1. For example, a test for serially correlated errors and an ARCH test can be performed as follows:

```
fitted_Canada <- VARshrink(Y, p = 2, type = "none", method = "ns")
```

Table 2: Information criteria (AIC and BIC) for model comparison.

	AIC.p=1	BIC.p=1	AIC.p=2	BIC.p=2	AIC.p=3	BIC.p=3
Ordinary least squares	467.6	515.7	444.2	530.4	452.3	576.2
Ridge regression	465.8	504.6	442.9	509.3	445.3	525.9
Nonparametric shrinkage	454.5	478.9	426.6	471.4	422.5	485.4
Full Bayes	457.0	497.2	444.2	508.2	445.3	523.8
Semi Bayes	533.2	577.5	574.1	656.1	475.0	553.4
K-fold CV	495.7	538.2	466.2	525.4	472.9	538.1

```
serial.test(fitted_Canada)
```

```
#>
#> Portmanteau Test (asymptotic)
#>
#> data: Residuals of VAR object fitted_Canada
#> Chi-squared = 176.31, df = 228, p-value = 0.9953
```

```
arch.test(fitted_Canada)
```

```
#>
#> ARCH (multivariate)
#>
#> data: Residuals of VAR object fitted_Canada
#> Chi-squared = 508.04, df = 500, p-value = 0.3921
```

We can inspect the summary results to assess the statistical significance of each covariate affecting the unemployment rate through the model coefficients:

```
summary_Canada <- summary(fitted_Canada)
print(summary_Canada$varresult[["U"]])
```

```
#>
#> Call:
#> VARshrink(y = Y, p = 2, type = "none", method = "ns")
#>
#> Residuals:
#>      Min       1Q   Median       3Q      Max
#> -0.84400 -0.16754 -0.00281  0.16971  0.72526
#>
#> Coefficients:
#>           Estimate Std. Error t value Pr(>|t|)
#> e.l1      -0.36506    0.03356 -10.877 < 2e-16 ***
#> prod.l1   -0.13332    0.03502  -3.807 0.000283 ***
#> rw.l1      0.03840    0.02940   1.306 0.195426
#> U.l1       0.07101    0.03193   2.224 0.029098 *
#> e.l2       0.01561    0.03265   0.478 0.633932
#> prod.l2   -0.04137    0.03533  -1.171 0.245312
#> rw.l2      0.09894    0.02930   3.377 0.001157 **
#> U.l2      -0.11933    0.03161  -3.775 0.000314 ***
#> ---
#> Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
#>
#> Residual standard error: 0.2937 on 76.32037 degrees of freedom
#> Multiple R-squared:  0.5625, Adjusted R-squared:  0.5414
#> F-statistic: 26.67 on 3.679632 and 76.32037 DF, p-value: 2.873e-13
```

It is worthwhile to examine the correlations among the noise variables to evaluate concurrent relationships between the time series variables within the residuals:

```
print(round(summary_Canada$corres, 3))
```

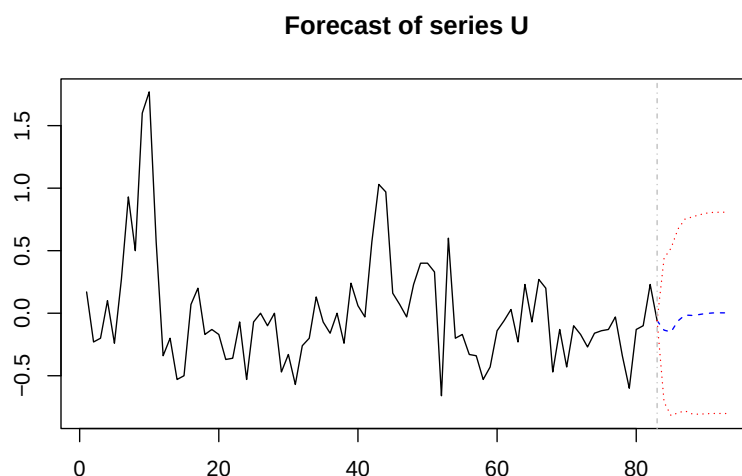


Figure 3: 10-step-ahead forecast of the differenced Canada time series using the VAR model estimated via the NS method with lag order $p = 2$.

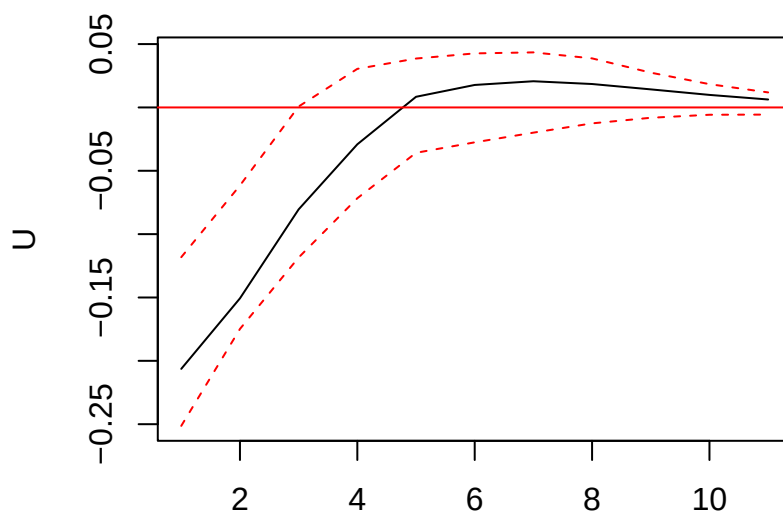
```
#>      e  prod  rw  U
#> e    1.000 -0.070 -0.039 -0.687
#> prod -0.070 1.000 0.086 0.018
#> rw   -0.039 0.086 1.000 0.207
#> U    -0.687 0.018 0.207 1.000
```

Time series forecasting can be performed, as illustrated in Figure 3.

```
plot(predict(fitted_Canada), names = "U")
```

Impulse responses can be computed from the same fitted object.

Orthogonal Impulse Response from e



95 % Bootstrap CI, 100 runs

4.2 Comparative simulation

We generated multivariate time series data of length $T = 20, 40, 80, 160$ from $K = 20$ dimensional VAR models of order $p = 1$. Note that for $T = 20$, the sample size $N = T - p = 19$ is smaller than the dimensionality $K = 20$. The VAR coefficient matrix $\mathbf{A}_1$ was specified with diagonal entries of 0.6 and zeros elsewhere, except for K entries randomly selected from the lower triangular part of the matrix.

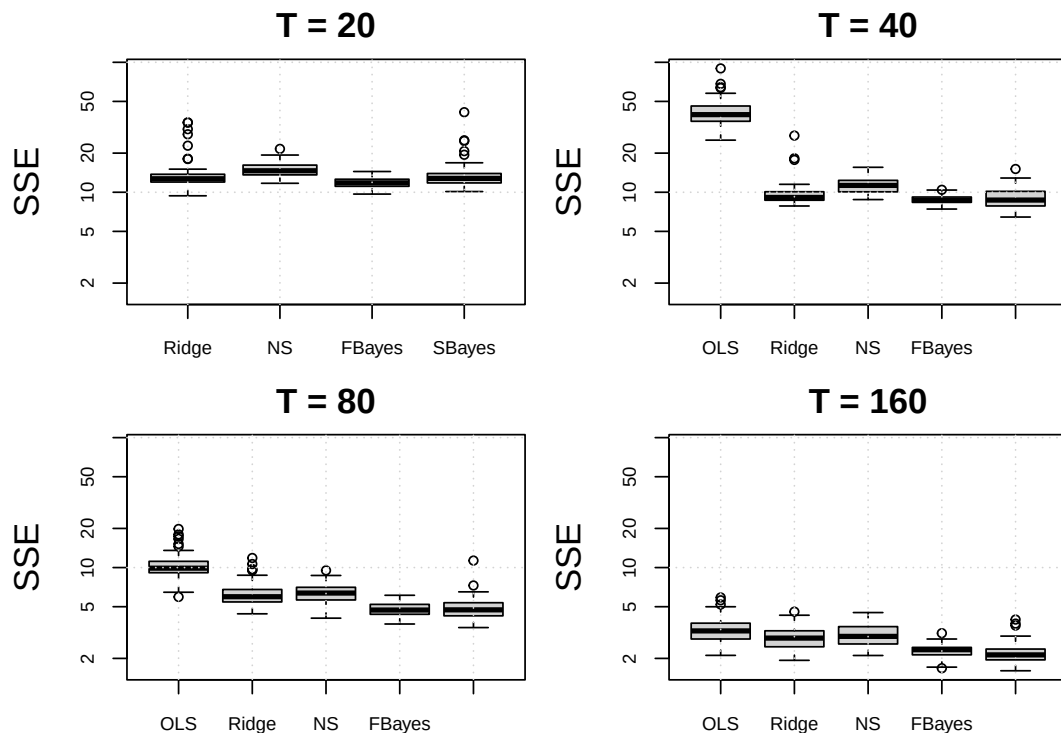


Figure 4: Comparison of mean SSEs across the four shrinkage methods and the OLS.

The nonzero off-diagonal entries were drawn uniformly at random from the intervals $[-1, -0.2]$ and $[0.2, 1]$. Since the coefficient matrix is lower triangular, the diagonal entries determine its eigenvalues, which lead to weak stationarity (Hamilton, 1994; Tsay, 2005). The noise ϵ_t was randomly sampled from the multivariate normal distribution whose covariance matrix has diagonal entries of 1 and off-diagonal entries of 0.5. The experiments were repeated 50 times. We compared the four shrinkage methods and, when $T - p > K$, the OLS baseline. The sum of squared errors (SSEs) of the estimated VAR coefficients was used as a measure of parameter estimation accuracy. The summary of the coefficient SSEs is shown in Figure 4. The figure shows that the full Bayesian and semiparametric Bayesian methods produced accurate estimates with low SSEs, particularly for larger values of T , as they correctly incorporated correlations among the noise variables.

Figure 5 presents the out-of-sample forecasting performance of the shrinkage methods and the OLS baseline when estimable. Figure 6 presents the computational costs of the same methods as a function of the sample size. The full Bayesian method required the largest computational cost, indicating that computationally efficient alternative methods are needed for high-dimensional time series analysis.

To assess scalability with respect to the number of VAR coefficients, we also varied the dimension K while holding $T = 160$ and $p = 1$ fixed. The number of autoregressive coefficients is $K^2 p$. As shown in Figure 7, computational costs increase much more rapidly with the number of VAR coefficients than with the sample size.

5 Discussion

The **R** ecosystem for multivariate time series has become increasingly specialized, as reflected in the CRAN Time Series Task View (Hyndman and Killick, 2026). Packages such as **vars** and **MTS** provide established tools for VAR estimation, forecasting, and post-estimation analysis. Other packages focus on modern extensions: **BigVAR** and **bigtime** emphasize penalized estimation for large multivariate systems (Nicholson et al., 2026; Wilms et al., 2023), while **BVAR**, **bayesianVARs**, **bvars**, and **bvartools** support Bayesian workflows involving hierarchical prior selection, MCMC estimation, and stochastic volatility (Giannone et al., 2015; Kuschnig and Vashold, 2021; Gruber et al., 2026; Primiceri, 2005; Krueger, 2015; Mohr, 2024). For structural Bayesian VARs, **bsvarSIGNs** provides tools for sign, zero, and narrative restrictions (Wang and Woźniak, 2025).

VARshrink is designed to complement these packages by bringing an OLS baseline and several shrinkage estimators into a common interface compatible with the widely used **vars** workflow. It currently includes multivariate ridge regression (Hoerl and Kennard, 1970; Golub et al., 1979), a Stein-type nonparametric shrinkage method (Opgen-Rhein and Strimmer, 2007), fully Bayesian methods

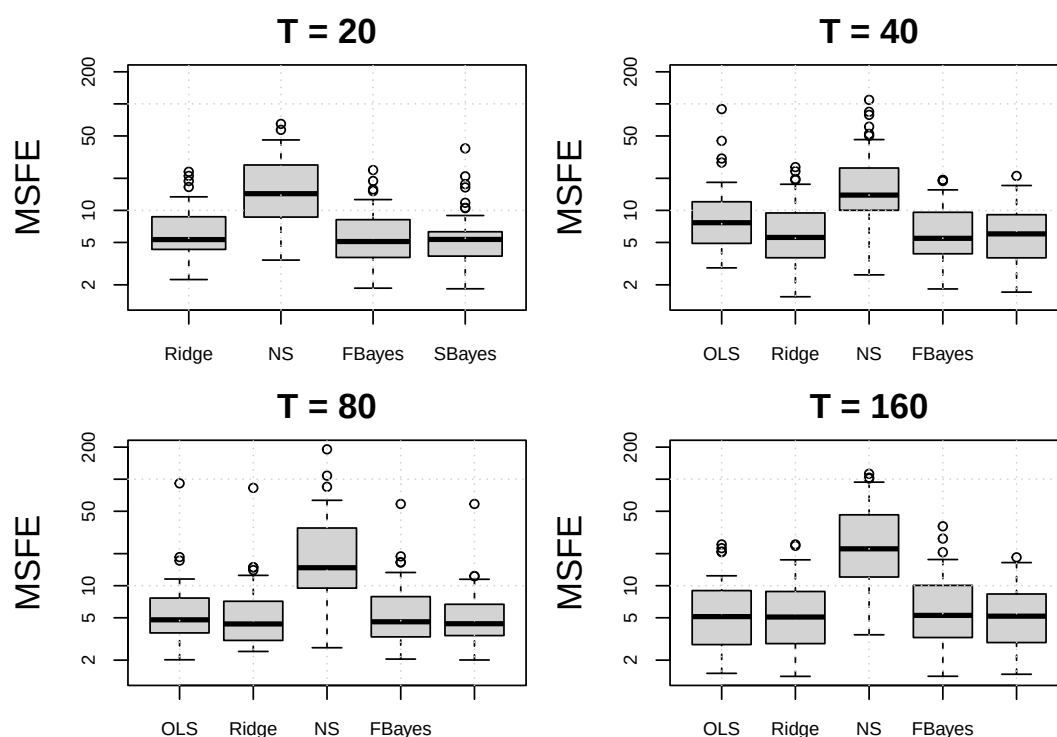


Figure 5: Comparison of 10-step out-of-sample MSFEs across the four shrinkage methods and the OLS baseline when estimable.

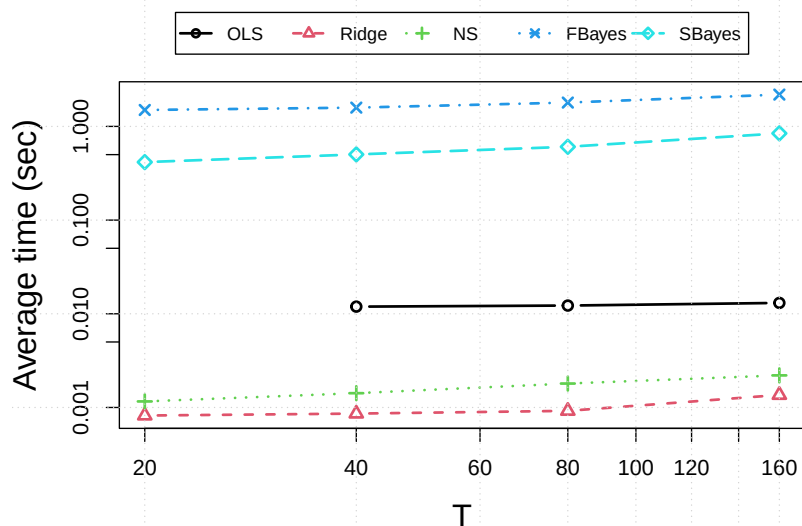


Figure 6: Comparison of computational costs across the four shrinkage methods and the OLS baseline when estimable as sample size increases.

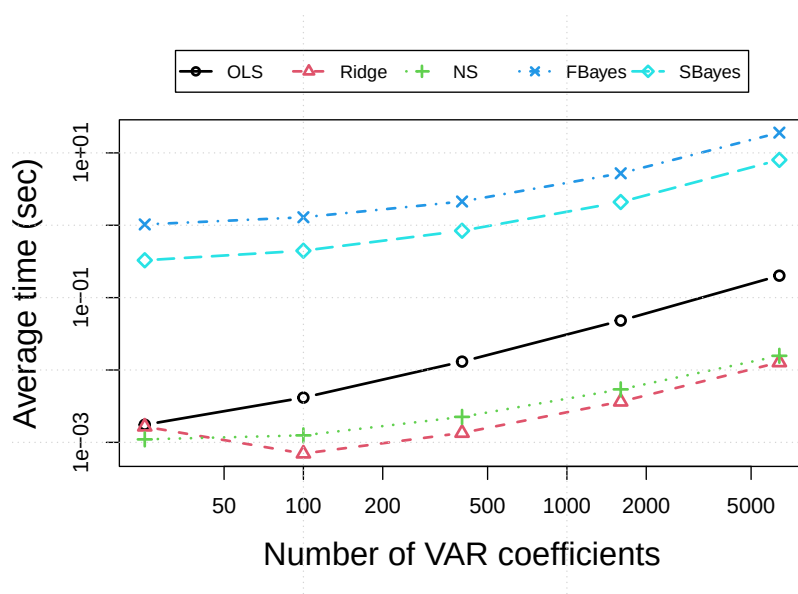


Figure 7: Comparison of computational costs across the four shrinkage methods and the OLS baseline as the number of VAR coefficients increases.

with natural conjugate and non-conjugate priors (Ni and Sun, 2005), and semiparametric Bayesian methods (Lee et al., 2016). This design allows users to compare OLS and alternative shrinkage estimators while retaining familiar tools for coefficient extraction, fitted values, residuals, forecasting, impulse responses, forecast error variance decompositions, diagnostic tests, and visualization.

The package is intended for applied researchers, students, and analysts who use the **vars** workflow but need more stable estimation than ordinary least squares can provide. Such settings arise when the number of variables is large relative to the number of observations, as in macroeconomic forecasting, financial systems, gene regulatory network analysis, and dynamic brain connectivity analysis (Harrison et al., 2003; Opgen-Rhein and Strimmer, 2007; Bańbura et al., 2010). In these applications, shrinkage is useful not only for forecasting but also for stabilizing coefficient estimates used to study dynamic relationships, partial correlation structures, impulse responses, and forecast error variance decompositions.

The structural analysis tools in **VARshrink** come from this interoperability with **vars**. Because **varshrinkest** objects inherit from **varest**, users can apply familiar post-estimation tools from **vars**. For impulse responses, uncertainty calculation is estimator-specific: bootstrap re-estimation can be costly for shrinkage estimators, whereas stored MCMC draws can be used by `irf()` to form Bayesian credible intervals when the full Bayesian method is fitted with `store_mcmc = TRUE`. Users whose main goal is advanced structural identification, stochastic volatility, or sign, zero, and narrative restrictions may therefore prefer packages designed specifically for those purposes.

Several limitations remain. **VARshrink** does not yet implement the full range of modern Bayesian shrinkage priors, such as Minnesota-type, global-local, or hierarchical priors used in macroeconomic forecasting (Doan et al., 1984; Litterman, 1986; Kastner and Huber, 2017; Giannone et al., 2015). It also inherits structural identification from **vars** rather than developing a separate structural VAR framework. Finally, fully Bayesian estimation provides richer posterior information but is more computationally expensive than ridge or nonparametric shrinkage, especially as the number of variables grows. These limitations define the current scope of the package and suggest future extensions, including additional prior structures, structured penalties, and closer integration with modern structural VAR workflows.

6 Conclusions

Shrinkage methods are useful for VAR models when the number of parameters is large relative to the sample size. The R package **VARshrink** provides an OLS baseline and ridge, nonparametric, fully Bayesian, and semiparametric shrinkage estimators through a unified interface connected to the familiar **vars** class system. This allows users to compare OLS with shrinkage methods and then apply standard VAR tools for forecasting, visualization, diagnostics, impulse responses, and forecast error variance decompositions.

VARshrink complements specialized software for penalized forecasting, Bayesian prior selection, stochastic volatility, and structural identification. Its main role is to provide a common workflow for stable VAR coefficient estimation in high-dimensional settings. Future development can extend this framework with additional prior structures, structured penalties, and richer structural analysis tools while preserving compatibility with established VAR workflows in **R**.

Computational details

The results in this manuscript were obtained using **R** 4.5.3 and **VARshrink** 0.5.0 on a MacBook running macOS 26.3.1, equipped with an Apple M4 Pro CPU and 24GB of memory.

Supplementary materials

varshrink-rjournal.R: R code to reproduce all the results in the manuscript.

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Conflict of interest

The authors have no competing interests to declare that are relevant to the content of this article.

Data availability

The Canada dataset was deposited by other researchers and is available in the **R** package **vars** at the CRAN repository, <https://CRAN.R-project.org/package=vars>.

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